Statistical & Numerical Methods Final Assignment

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(Date: July 15, 2026)

1 Lift Waiting Location

The code solving this, with documentation, is attached as q1.py, the result it produces is floor 4.

1.1 Bonus Calculation

To find the optimal resting floor we'll start by analyzing the paths it takes, focusing on a single person. Through all the assumptions we can simplify the model to:

1. Person enters on floor 0.
2. Person go to floor X.
3. Person then goes to 3 random floors.
4. From the final floor he reaches, the person goes down to 0 and leaves.

in each movement step the elevator performs

1. Elevator goes to origin floor from its resting point.
2. Elevator goes to destination floor from the origin floor.
3. Elevator goes to the resting point from the destination floor.

Each ride's cost calculation is

$$C = |rest - origin| + |dest - origin| + |rest - dest| \tag{1}$$

we can notice that the middle element doesn't depend on the rest point so we can simply ignore it. Therefore the goal we want to minimize is

$$C = E\left(|rest - origin|\right) + E\left(|rest - dest|\right) \tag{2}$$

but from the symmetry of a sufficiently large random selection of people and trips we can simplify this to

$$C = 2E\left(|rest - X|\right) \tag{3}$$

where X is a floor. To this we add the trips as the mapping

$$\{0, f_1, f_2, f_3, f_4\} \to \{f_1, f_2, f_3, f_4, 0\} \tag{4}$$

we notice that the floor 0 appears twice and because the person travels from/to there in two separate trips, so when working with the floor numbers, we'll need to weight trips to/from the floor 0 at double the weight of the others. We can now calculate

$$E\left(|rest - X|\right) = \frac{rest}{5} + \frac{4}{5} \cdot \frac{1}{10} \sum_{X=1}^{10} |rest - X| \tag{5}$$

we take the derivative to find the optimum

$$\frac{\mathrm{d}E}{\mathrm{d}rest} = \frac{1}{5} + \frac{4}{50} \sum_{X=1}^{10} \operatorname{sgn}\left(rest - X\right) \tag{6}$$

and

$$\frac{4}{50} \sum_{X=1}^{10} \operatorname{sgn}\left(rest - X\right) + \frac{1}{5} = 0 \tag{7}$$

Which is a pain to solve so we turn to tricks. If we assume there are k floors below the resting point, this means the expression inside the sum will be $+1$ k times, and -1 $10 - k - 1$ times. Therefore

$$D = \frac{4}{50} \sum_{X=1}^{10} \operatorname{sgn}\left(rest - X\right) + \frac{1}{5} = \frac{1}{5} + \frac{4}{50}\left(k + k - 9\right) = \frac{10 - 36 + 8k}{50} = \frac{8k - 26}{50} \tag{8}$$

plugging in the different possible values for k

$$D(0) = -\frac{26}{50} \qquad D(1) = -\frac{18}{50} \qquad D(2) = -\frac{10}{50} \qquad D(3) = -\frac{2}{50} \qquad D(4) = \frac{6}{50} \tag{9}$$

so our two suspect floors are 3 and 4. We look at the values at the floors:

$$E\left(|3 - X|\right) = \frac{3}{5} + \frac{4}{50} \sum_{X=1}^{10} |3 - X| = \text{calculator} = \frac{77}{25} \tag{10}$$

$$E\left(|4 - X|\right) = \frac{4}{5} + \frac{4}{50} \sum_{X=1}^{10} |4 - X| = \text{calculator} = \frac{74}{25} \tag{11}$$

so floor 4 gives a lower expected value, meaning it is our optimal floor.

2 Peaches and Oranges

Since we have no information on any distributions or chanves, we'll treat the fruit counts as being distributed poisson.

2.1 Oranges in the 2376 day

We can use the total number of peaches counted to find the expected value of peaches per day

$$\lambda = 10500/7 = 1500 \tag{12}$$

so the number of oranges is

$$N = 2376 - \lambda = 876 \tag{13}$$

The number of fruit counted the specific day is exact so has no variance. The number of peaches has variance which comes both from the variance on the actual count of peaches and from the daily count fluctuating around the average we calculate.

$$\sigma_{peach,week} = \sqrt{10500} \sim 102.5 \tag{14}$$

but since we divided it by 7 to look at each day

$$\sigma_{peach,day} = \frac{102.5}{7} \sim 14.6 \tag{15}$$

then the variance due to the actual daily number not being the exact average

$$\sigma_{day} = \sqrt{1500} \sim 38.7 \tag{16}$$

so the number of oranges that day is

$$N = 876 \pm 14.6 \pm 38.7 \tag{17}$$

and since those are independent the total error is

$$\sigma_{total} = \sqrt{(14.6)^2 + (38.7)^2} \sim 41.4 \tag{18}$$

so

$$N = 876 \pm 41.4 \tag{19}$$

2.2 Expected Daily Count

This time the count of total fruit is also uncertain. We treat it as a single measurement from a poisson distribution meaning

$$\lambda_{fruit} = 2376, \qquad \sigma_{fruit} \sim 48.7 \tag{20}$$

and as for the count of peaches, this time we don't have the second source of uncertainty, since we don't need to consider whether the specific day was average or not, since we're looking at the average itself, so

$$\lambda_{peaches} = 1500, \qquad \sigma_{peaches} = 14.6 \tag{21}$$

which gives us

$$N = 876 \pm 48.7 \pm 14.6 \tag{22}$$

and again summing independent errors

$$\sigma = \sqrt{48.7^2 + 14.6^2} \sim 50.8 \tag{23}$$

so

$$N = 876 \pm 50.8 \tag{24}$$

3 Resistors

Since the resistors are connected in parallel, we'll find the equivalent total resistance using

$$\frac{1}{R_{tot}} = \sum \frac{1}{R} \implies R_{tot} = \frac{R_1 R_2 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3} \tag{25}$$

giving us

$$R_{tot} = 3.3\text{k}\Omega \tag{26}$$

3.1 Total error

The error on the total resistance will have to be calculated using error propagation

$$\sigma^2 = \left(\frac{\partial R_{tot}}{\partial R_1}\right)^2 \sigma_{R1}^2 + \left(\frac{\partial R_{tot}}{\partial R_2}\right)^2 \sigma_{R2}^2 + \left(\frac{\partial R_{tot}}{\partial R_3}\right)^2 \sigma_{R3}^2 \tag{27}$$

$$= 3\left(\frac{R_2 R_3 (R_1 R_2 + R_1 R_3 + R_2 R_3) - R_1 R_2 R_3 (R_2 + R_3)}{(R_1 R_2 + R_1 R_3 + R_2 R_3)^2}\right)^2 \sigma_{R1}^2 \tag{28}$$

$$= 3\left(\frac{R_1 R_2^2 R_3 + R_1 R_2 R_3^2 + R_2^2 R_3^2 - R_1 R_2^2 R_3 - R_1 R_2 R_3^2}{(R_1 R_2 + R_1 R_3 + R_2 R_3)^2}\right)^2 \sigma_{R1}^2 \tag{29}$$

$$= 3\left(\frac{R^4}{9R^4}\right)^2 \sigma_R^2 = \frac{1}{27}\sigma_R^2 \tag{30}$$

$$= \frac{1}{27}(0.05 \cdot 10\text{k}\Omega)^2 \tag{31}$$

$$= 9259\Omega^2 \tag{32}$$

so

$$\sigma = 96\Omega \tag{33}$$

where we used the fact the resistors are idential several times.

3.2 A single resistor

This time we only have a single source of error.

$$\sigma = 0.05 \cdot 3.3\text{k}\Omega = 165\Omega \tag{34}$$

3.3 in series instead

To get the same voltage we need the same resistance. The total resistance we found was 3.3kΩ so we need three resistors each 1.1kΩ. The equation for total resistance is

$$R_{tot} = R_1 + R_2 + R_3 \tag{35}$$

so

$$\sigma^2 = 3\left(\frac{\partial R_{tot}}{\partial R}\right)^2 \sigma_R^2 = 3\sigma_R^2 = 3 \cdot (0.05 \cdot 1.1\text{k}\Omega)^2 = 9075\Omega^2 \tag{36}$$

so

$$\sigma = 95\Omega \tag{37}$$

3.4 Different resistances vs the same

Yes there is a difference. Since they are independent elements, we sum the squares of the standard deviations then take the square root. If we look at the limit where one resistor has all the resistance and the others have resistance 0, the error is the same as the error on the individual resistor. Moving resistance to another resistor takes the value out from the square in the error calculation and moves it to a multiplication inside the root, which reduces it.

4 nonlinear system

The python code solving this with documentation is attached as several python files, which will be explained fully in a section of this answer. For all the tests I used the reduced chi-square test.

4.1 Explanation of the code

4.2 Representative measurements

The results from running the code for the representative measurements are

Model	χ^2	Reduced χ^2	p-value
Y_1	84.6	6.5	1.5×10^{-12}
Y_2	163.6	12.6	3.7×10^{-28}
Y_3	26.0	2.0	0.017

Table 1 – The chi-square, reduced chi-square, and p-value for the best fit for each model to the representative measurement dataset.

with the optimal parameters being

Model	A	B	C	D
Y_1	128.56	0.22	-186.57	-133.43
Y_2	-175.27	Inf.	0.4683	-67.60
Y_3	-1.5	28.04	-100.47	-159.35

Table 2 – Optimal parameters found using the python code. Note the infinity for B in model Y_2, which results in the optimizer going to arbitrarily large values, when it's allowed to.

The benefits of each hypothesis is as follows:

Y_1 a sum of two sine functions with double the frequency. From plotting the representative data, it indeed looks like the data might be some kind of oscillatory behavior, so the choice makes sense. If we expect the signal to be a sine wave with sine-like disturbances, this would capture it.

Y_2 a decaying sine. The benefit of this is that it allows the ability to capture a signal whose amplitude decays over time. The problem is that if the system does not decay, this will be unable to capture the behavior.

Y_3 a cubic polynomial. The benefits are that this is a much easier hypothesis to use for calculations and to test, and it fits the dataset in this part very well. The problem is that it assumes the behavior is of a cubic equation, which relies very strongly on an assumption the data will conform to that assumption beyond the measurements in the data.

The initial conclusion we can reach from this data is that models Y_1 and Y_2 can be ruled out for this dataset. Their high reduced chi-square combined with a miniscule p-value mean that the difference between the measurements and the model's prediction are both large and highly unlikely to be statistical errors. Y_3 also has a fairly high chi-square at 2 meaning the fit is still not great, but it's within workable limits.

4.3 Real Measurement

After looking at the raw data in a plot, I noticed that it can be broken up into two zones:

1. A background measurement
2. The signal region

and it appears that the noise in both regions is different. The signal measurement therefore itself introduces noise which we can't account for just with the background section. Looking at a histogram of the values in the background measurement I saw it looked fairly gaussian so I ran a chi-square test which resulted in

$$\chi^2/dof = 2.15, \qquad p - value = 0.018 \tag{38}$$

which is not a great result, but given I have no other information to work off of I'll accept an assumption the noise is gaussian in the background measurement with the following parameters. A further analysis that could be made is whether the noise is somehow self-correlated, i.e that not all measurements in the noise really are random, which if removed could be characterized better by a gaussian, but this is out of scope.

$$\mu = -2.688 \pm 0.132, \qquad \sigma = 3.68 \tag{39}$$

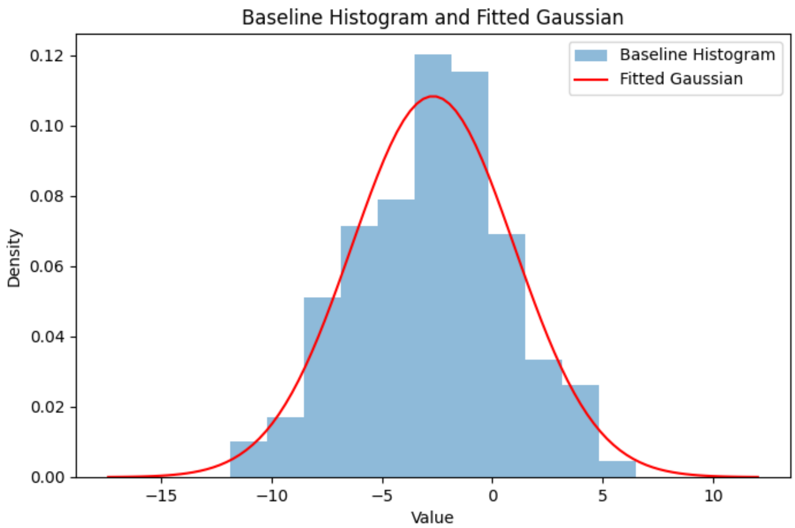

Figure 1 – Histogram of the values recorded in the section before the signal begins. A gaussian curve fit is presented on top.

To analyze the noise in the signal region, I have to first isolate the signal. Since I don't know which of the hypothesis are correct and I don't want to rush to the next part, I isolated it by performing a rolling average of the values, with a window of 20 measurements. This window was significantly smaller than the oscillations in the signal so it did not skew the signal too much.

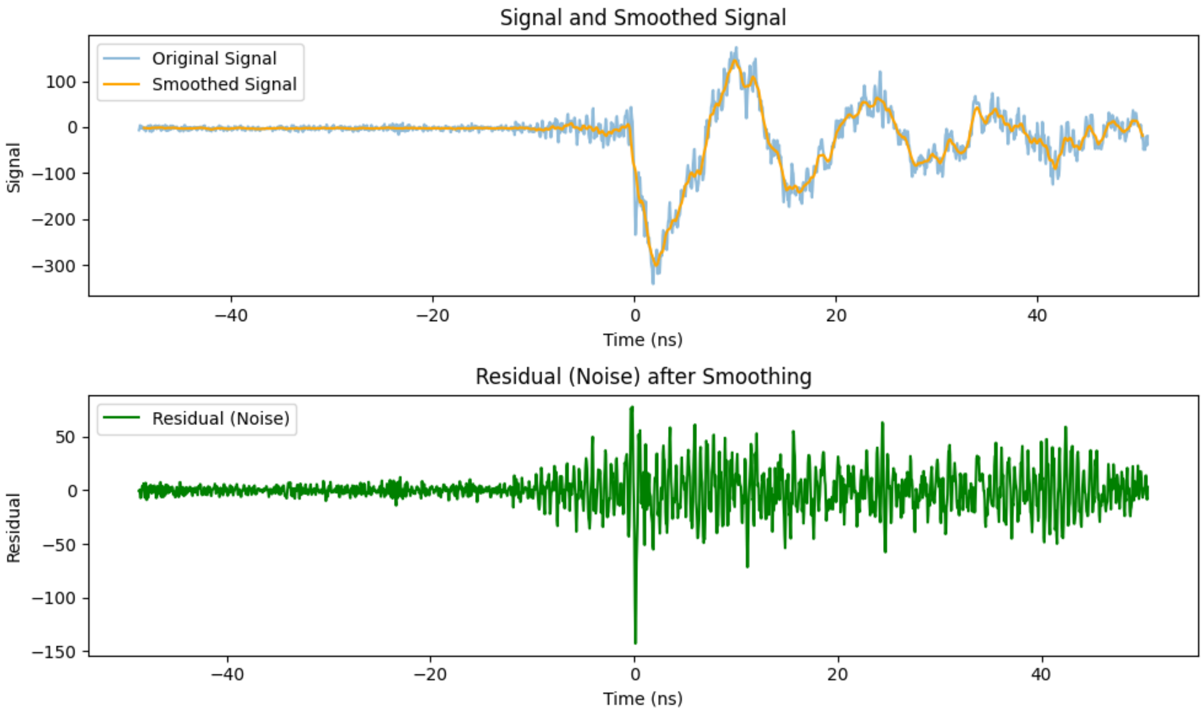

Figure 2 – Top panel: The raw data in blue and the smoothed out signal in yellow. Bottom panel: The residual, i.e the raw data with the smoothed out signal subtracted.

I then subtracted the cleaned signal from the raw data and found the noise in the signal region. After plotting a histogram of it, it also appeared quite gaussian, so I ran another chi-square test:

$$\chi^2/dof = 0.81, \qquad p - value = 0.615 \tag{40}$$

which indicates that this time the noise is actually somewhat well described as gaussian:

$$\mu = -0.277, \qquad \sigma = 22.438 \tag{41}$$

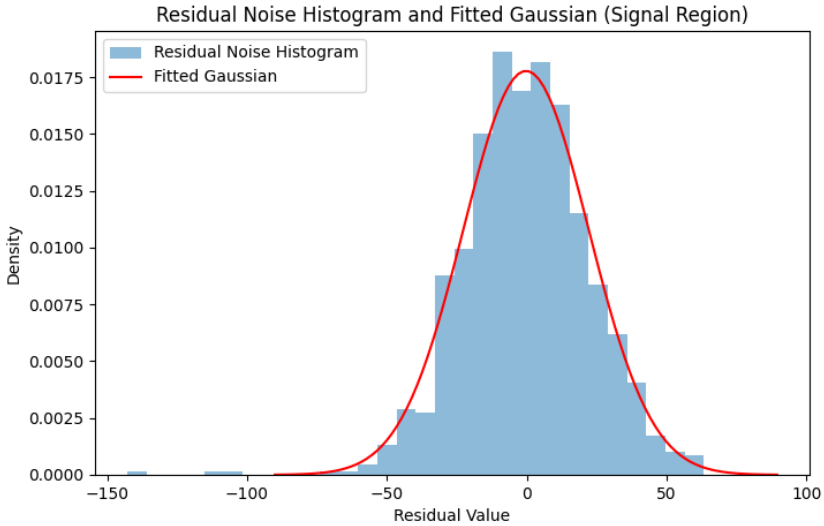

Figure 3 – Histogram of the values of the residual noise with a fitted gaussian.

next to characterize the signal I split the signal into bins of 0.5ns, and for each of those I calculated the mean, which is the total count N, then from it subtracted the baseline noise's mean which acts as B, giving me $S = N - B$. I then calculated the error from its own standard deviation divided by the number of counts in the bin combined with the erro on B.

$$S_{\text{min}} = -295 \pm 7 \quad \text{at } t = 2.11\text{ns}, \qquad S_{\text{max}} = 154 \pm 4 \quad \text{at } t = 10.11\text{ns} \tag{42}$$

with significance

$$seg_{\text{min}} = 42, \qquad seg_{\text{max}} = 36 \tag{43}$$

meaning the signal is highly significant in comparison to the noise.

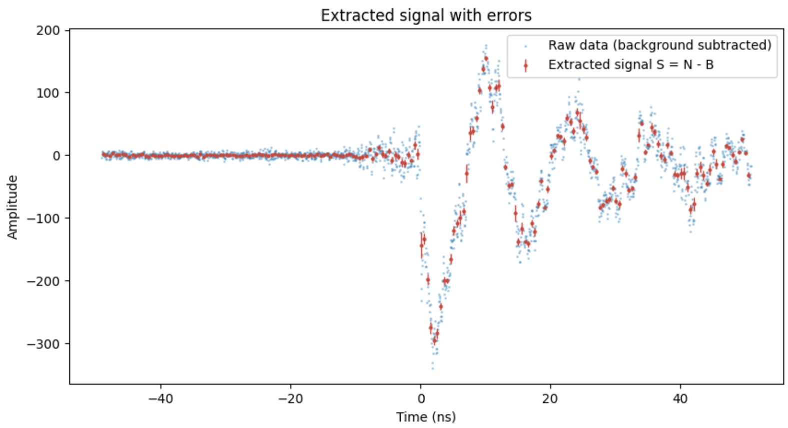

Figure 4 – The raw data in blue with the extracted signal, with errorbars, on top in red.

4.4 Simulation

In this part all that's needed is to load and properly process the data of the simulation then run the same tests as I did for the other three hypothesis. A note on the data processing, the simulation's times are different from the others, so the times needed to be shifted. Instead of trying to manually find the right shift, I found the correct shift with the same optimization technique used for the other parameters in the question, with minimizing the chi-square between the simulation and the measured data. I decided to re-fit the hypothesis Y_i to the real measurement data from part b, to give each hypothesis its best shot. This resulted in the follwoing

Hypothesis	χ^2/dof
Y_1	89.98
Y_2	27.92
Y_3	221.20
Simulation	118.03

Table 3 – The reduced chi squared for each hypothesis. The p value is not presented because it is so small it underflows the precision of floating point numbers, meaning it's basically 0 for ever fit.

Since the p value is effectively 0 for all of the fits, but none of the reduced chi-square tests give a good result, every hypothesis is rejected as a *good* fit. But, since we were asked for the best, rather than what's good, the best fit is the one with the lowest chi-squared, that being Y_2. Y_3 is completely rejected because it can't capture the behavior of the data at all, and Y_1 is rejected because it's simply worse. The simulation too is rejected for this reason.